



جامعة الملك فهد للبترول والمعادن
King Fahd University of Petroleum & Minerals

Feedback Circuits Note (01)

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- ☐ **Feedback Fundamentals**
- ☐ **Negative Feedback Topologies**
- ☐ **Negative Feedback: Benefits and Tradeoffs**
 - ☐ **Gain Robustness**
 - ☐ **Bandwidth Extension**
 - ☐ **Internal Disturbance/Noise Suppression**
 - ☐ **Lower In-Loop Distortion**
 - ☐ **Impedance Control**
- ☐ **Resistance Derivations**

Feedback Fundamentals (1/4)

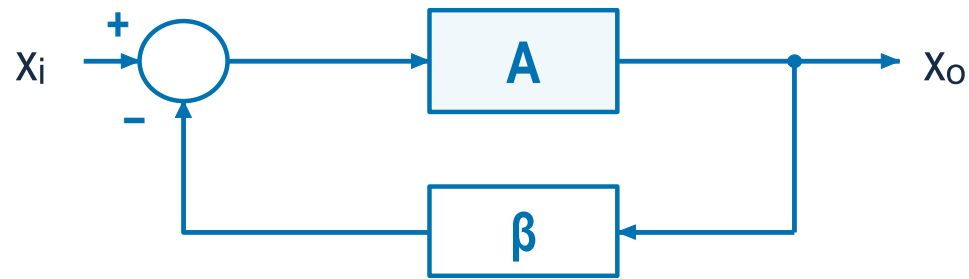


Feedback Circuits

- Feedback circuits are classified according to the polarity of the returned signal.

Negative feedback

The feedback signal is subtracted from the input.



Closed-loop gain

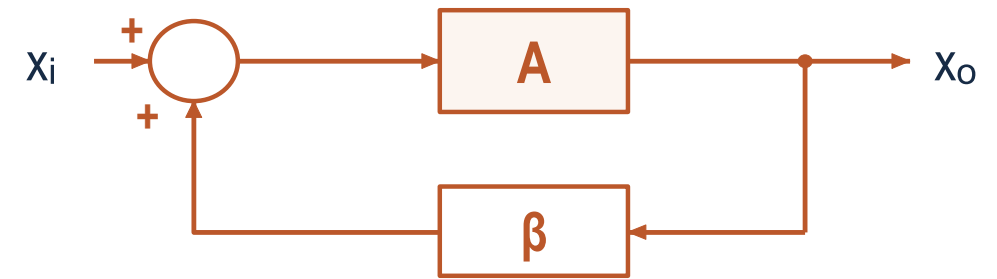
$$A / (1 + A\beta)$$

Application: amplifier design



Positive feedback

The feedback signal is added to the input.



Closed-loop gain

$$A / (1 - A\beta)$$

Application: signal and waveform generation



Square



Triangular



Sinusoidal

Feedback Fundamentals (2/4)



❑ Negative Feedback: Definitions and Gain

- ❑ The closed-loop gain is derived from the error signal and the feedback factor.

Signal and gain definitions

x_i	Input signal
x_o	Output signal
x_f	Feedback signal
$x_e = x_i - x_f$	Error signal
$A = x_o / x_e$	Open-loop gain
$A_f = x_o / x_i$	Closed-loop gain
$\beta = x_f / x_o$	Feedback factor

Closed-loop gain derivation

$$x_o = A \quad x_e = A(x_i - x_f)$$

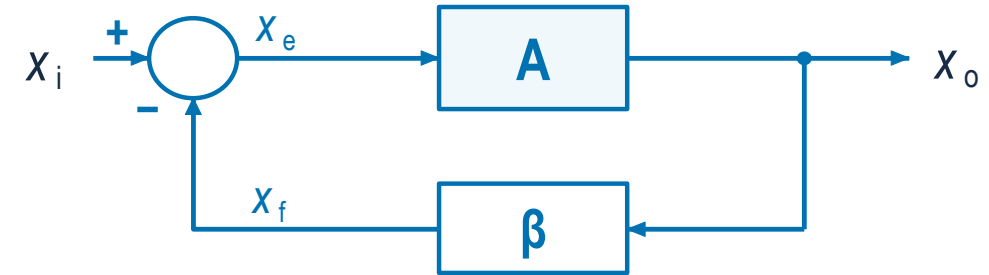
$$x_f = \beta \quad x_o$$

$$x_o = A(x_i - \beta x_o)$$

$$x_o(1 + A\beta) = A x_i$$

$A\beta$ denotes the loop gain.

Negative-feedback model



Negative feedback

$$A_f = \frac{x_o}{x_i} = \frac{A}{1 + A\beta}$$

Closed-loop gain = Forward gain / (1 + Loop gain)



Note: Positive-feedback counterpart

$$A_f = \frac{A}{1 - A\beta}$$

Feedback Fundamentals (3/4)

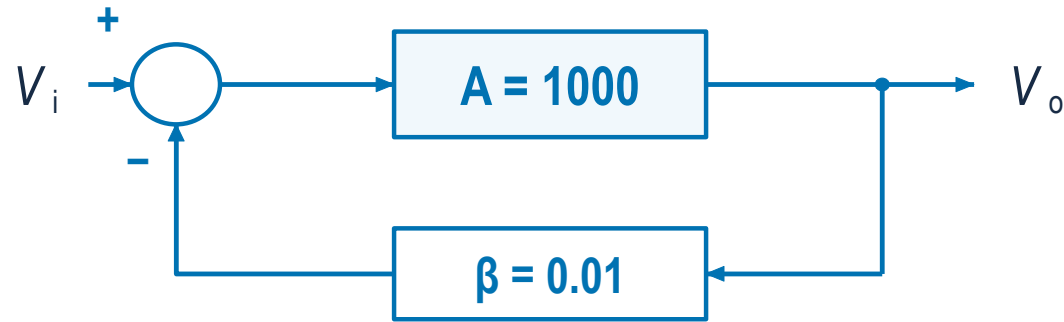


❑ Negative Feedback: Definitions and Gain

- ❑ The same amplifier is compared with and without a negative-feedback network.

With negative feedback

Feedback factor $\beta = 0.01$



$$A_f = \frac{V_o}{V_i} = \frac{A}{1 + A\beta}$$

$$A_f = 1000 / 11 \approx \mathbf{90.9}$$

Loop gain: $A\beta = 10$

Feedback denominator: $1 + A\beta = 11$

The gain trade-off: The reduced gain motivates analysis of feedback advantages and limitations.

Without feedback

Open-loop operation



$$A = \frac{V_o}{V_i} = 1000 \quad \text{Open-loop voltage gain}$$

Feedback Fundamentals (4/4)



□ Example → Non-Inverting Amplifier with Negative Feedback

- A high loop gain allows the resistor ratio to define the closed-loop gain.

Feedback factor and gain

$$A_f = \frac{V_o}{V_i} = \frac{A}{1 + A\beta}$$

$$\beta = \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2}$$

$$A_f = 1 / [1/A + R_1 / (R_1 + R_2)]$$

For $A\beta \gg 1$, the gain simplifies to

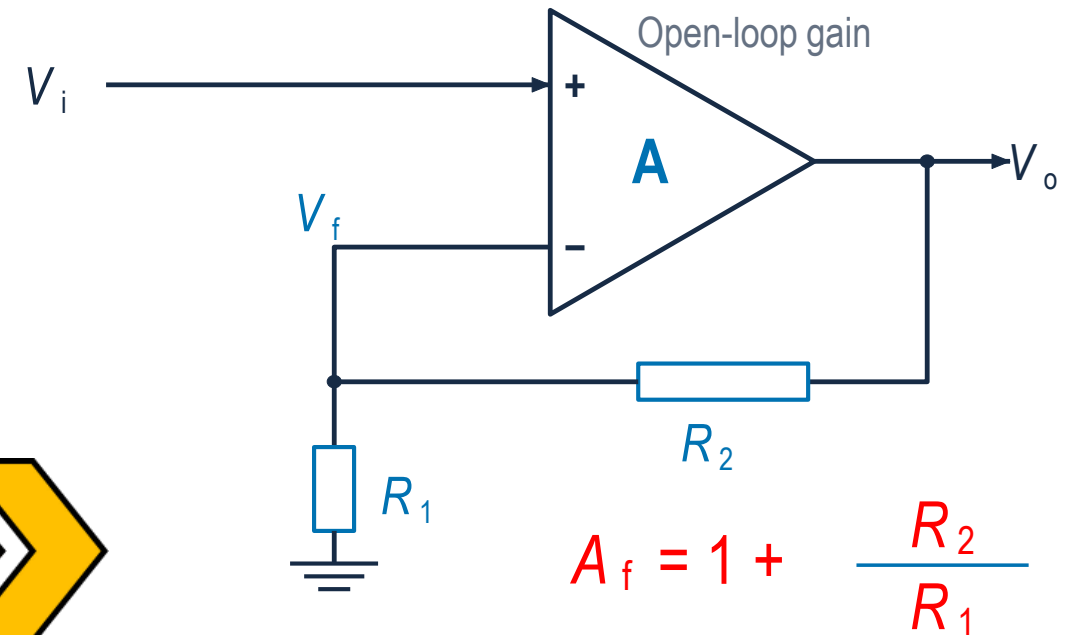
$$A_f \approx 1 + \frac{R_2}{R_1}$$



Gain robustness

- The resistor ratio sets the nominal closed-loop gain.
- A high loop gain reduces sensitivity to process and temperature variations in **A**.

Non-inverting topology



Illustrative open-loop gain variation

$A: 1000 \rightarrow 500$ (-50%)



- ❑ Feedback Fundamentals
- ❑ **Negative Feedback Topologies**
- ❑ **Negative Feedback: Benefits and Tradeoffs**
 - ❑ Gain Robustness
 - ❑ Bandwidth Extension
 - ❑ Internal Disturbance/Noise Suppression
 - ❑ Lower In-Loop Distortion
 - ❑ Impedance Control
- ❑ Resistance Derivations

Negative Feedback Topologies (1/8)



- ❑ Each topology is named by its input mixing and output sampling connections.

No.	Topology (input–output)	Input mixing	Output sampling
1	Series–series	Series	Series
2	Series–shunt	Series	Shunt
3	Shunt–shunt	Shunt	Shunt
4	Shunt–series	Shunt	Series

Input → mixing

Series mixing subtracts feedback **voltage**.

Shunt mixing subtracts feedback **current**.

Output → sampling

Shunt sampling senses output **voltage**.

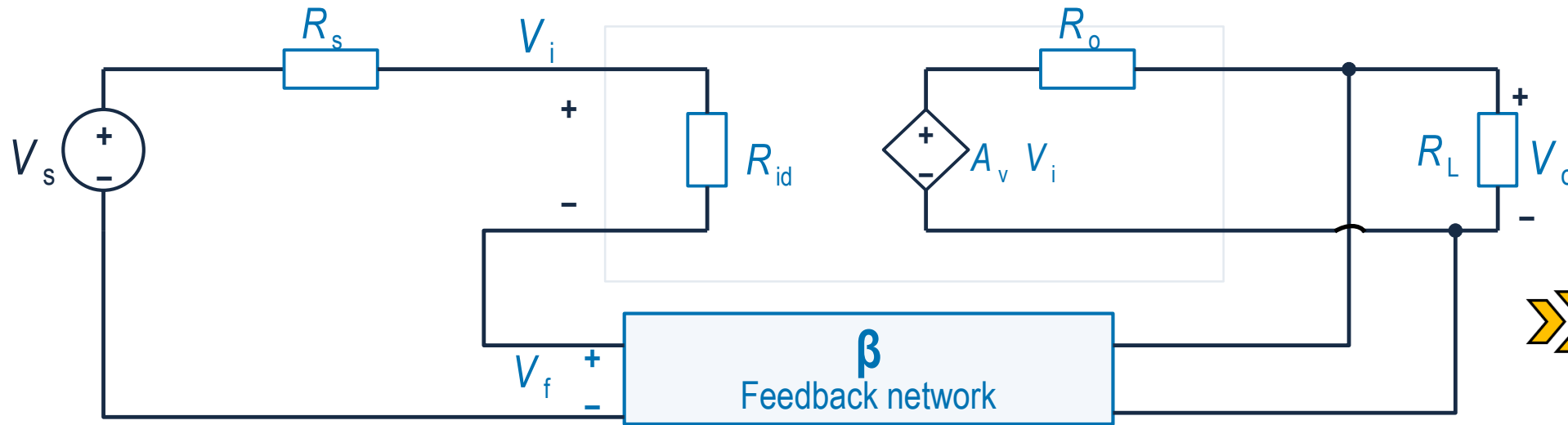
Series sampling senses output **current**.

Negative Feedback Topologies (2/8)



❑ Voltage Amplifier: **Series–Shunt Feedback**

- ❑ Voltage mixing at the input and voltage sampling at the output.



$$A_v = \frac{V_o}{V_i}$$
$$A_{vs} = \frac{V_o}{V_s}$$

Input mixing

Series voltage mixing

Input resistance increases

Output sampling

Shunt voltage sampling

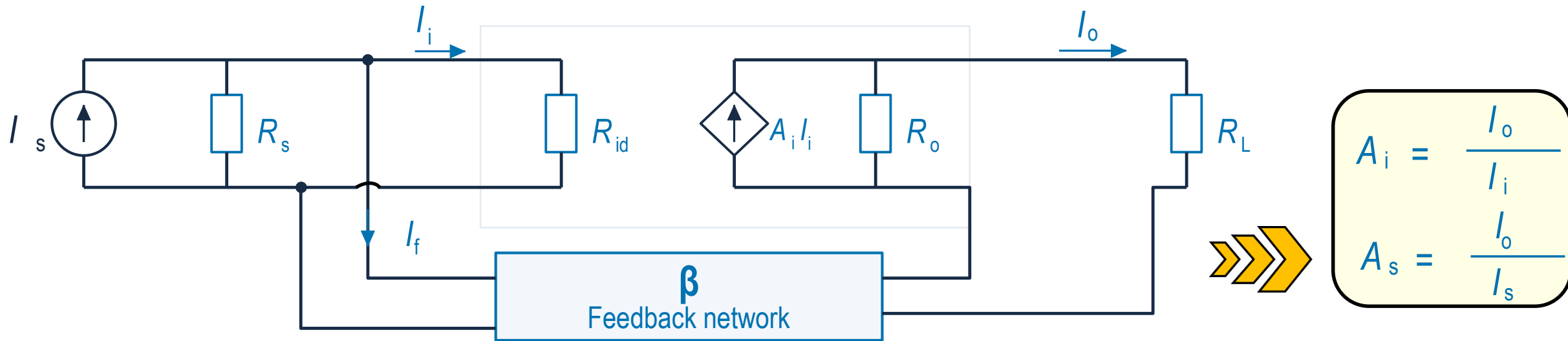
Output resistance decreases

Negative Feedback Topologies (3/8)



❑ Current Amplifier: **Shunt–Series Feedback**

- ❑ Current mixing at the input and current sampling at the output.



Input mixing

Shunt current mixing

Input resistance decreases

Output sampling

Series current sampling

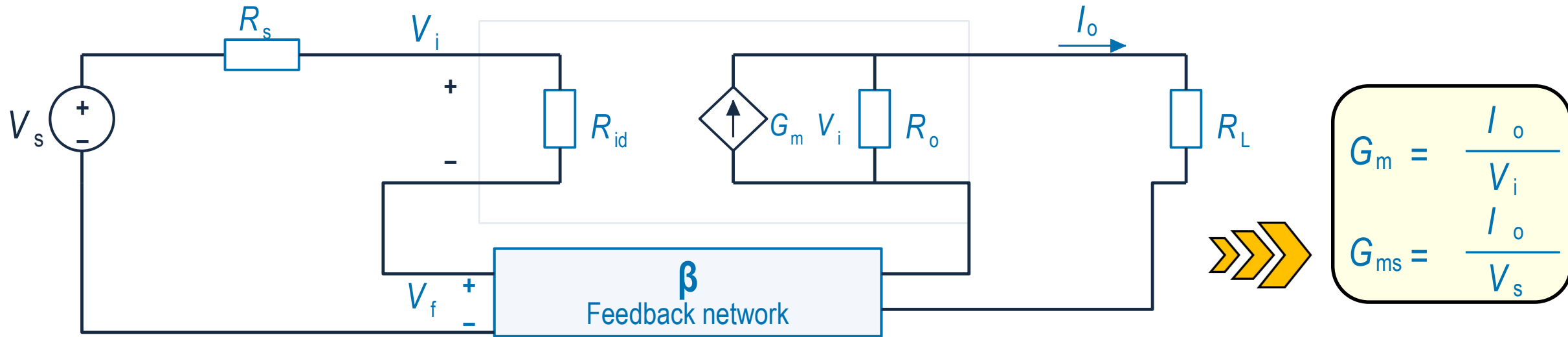
Output resistance increases

Negative Feedback Topologies (4/8)



❑ Transconductance: **Series–Series Feedback**

❑ Voltage input and current output define the transconductance amplifier.



Input mixing

Series voltage mixing

Input resistance increases

Output sampling

Series current sampling

Output resistance increases

-
- The diagram illustrates a feedback amplifier model. It features a current source I_s in parallel with a resistor R_s . This combination is connected to a node that splits into two parallel branches: one with resistor R_{id} and another with a dependent current source $R_m I_i$ in parallel with resistor R_o . The input current I_i flows into the R_{id} branch, and the feedback current I_f flows out of the $R_m I_i$ branch. The output voltage V_o is taken across a load resistor R_L . A feedback network block, labeled with the feedback factor β , is connected between the output and the input node. A large yellow arrow on the right indicates the signal flow.

$$R_m = \frac{V_o}{I_i}$$
$$R_{ms} = \frac{V_o}{I_s}$$

Output sampling

Shunt voltage sampling

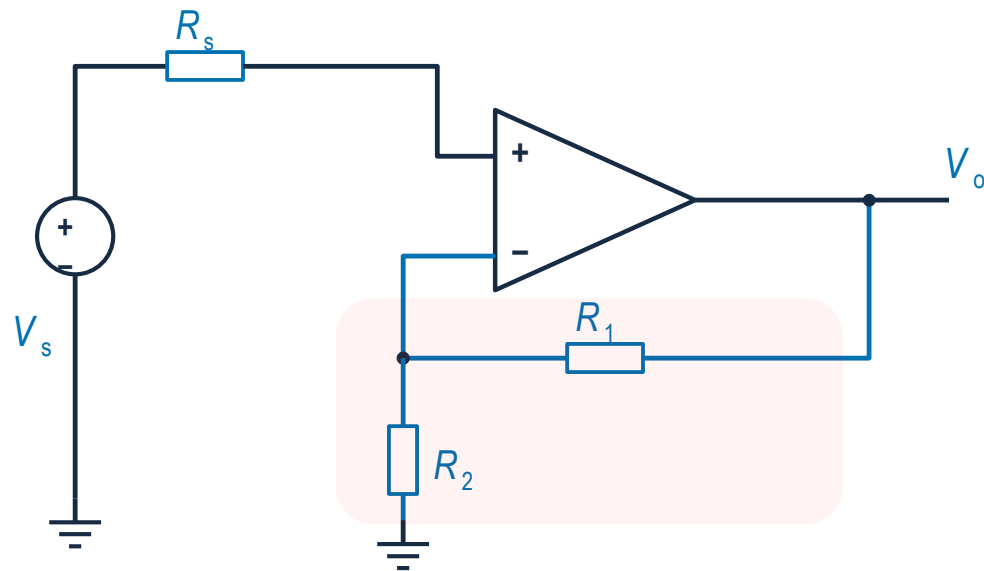
Output resistance decreases

Negative Feedback Topologies (6/8)



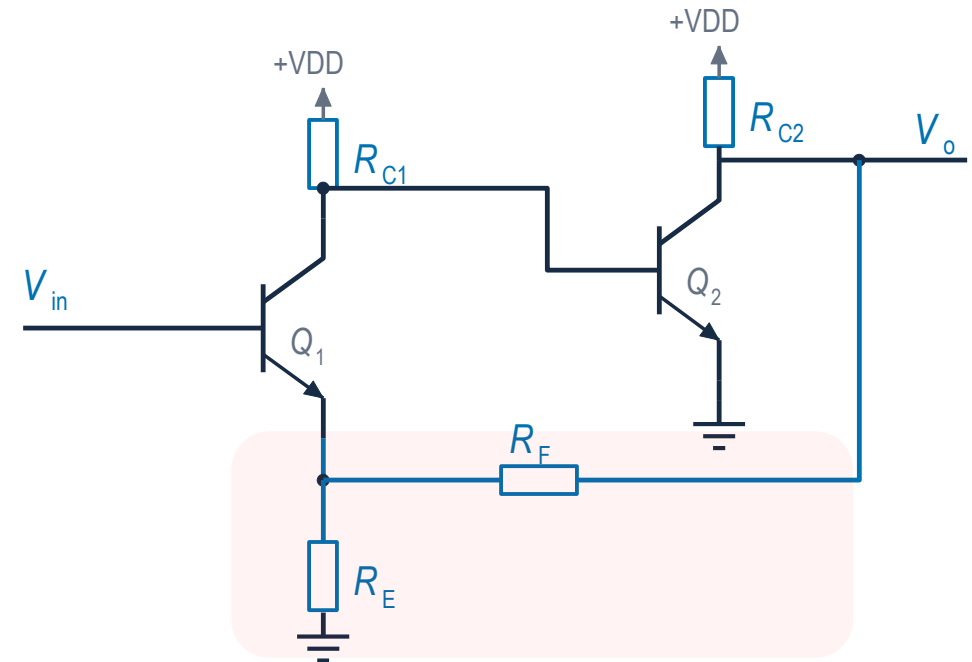
□ Circuit Examples

Series-Shunt



Non-inverting amplifier circuit

Series-Shunt



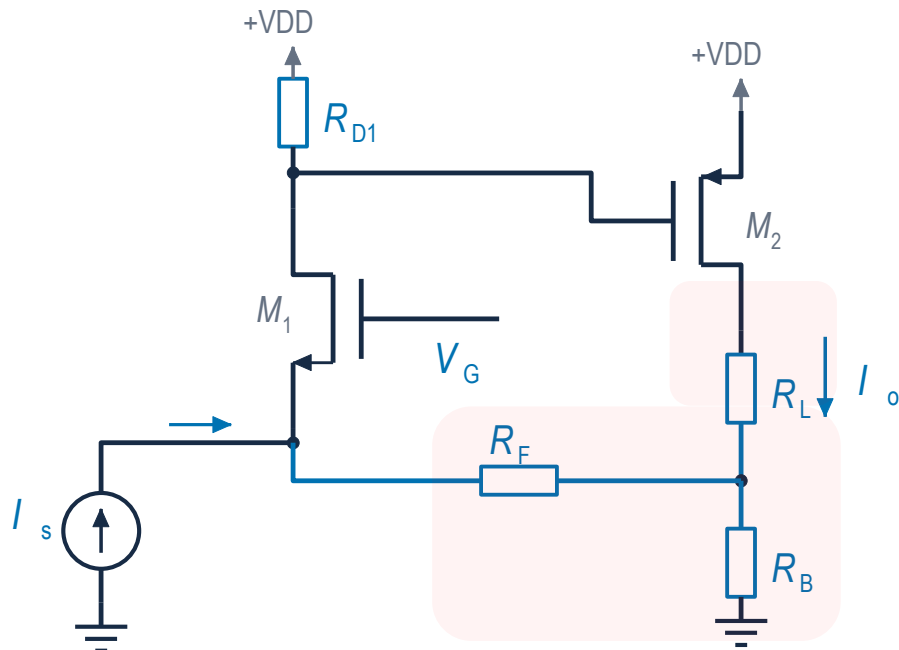
Two-stage BJT amplifier circuit

Negative Feedback Topologies (7/8)



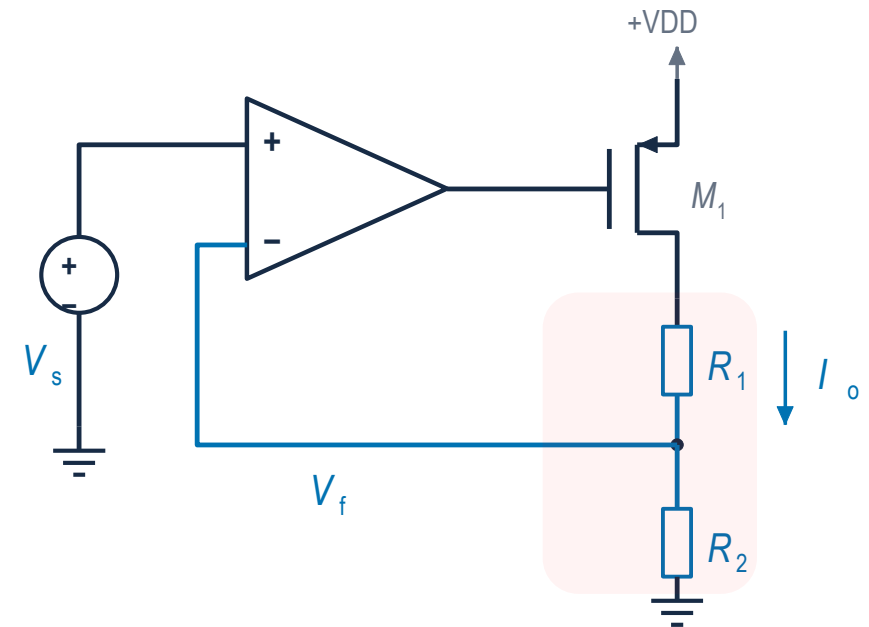
□ Circuit Examples

Shunt-Series



MOS amplifier circuit

Series-Series



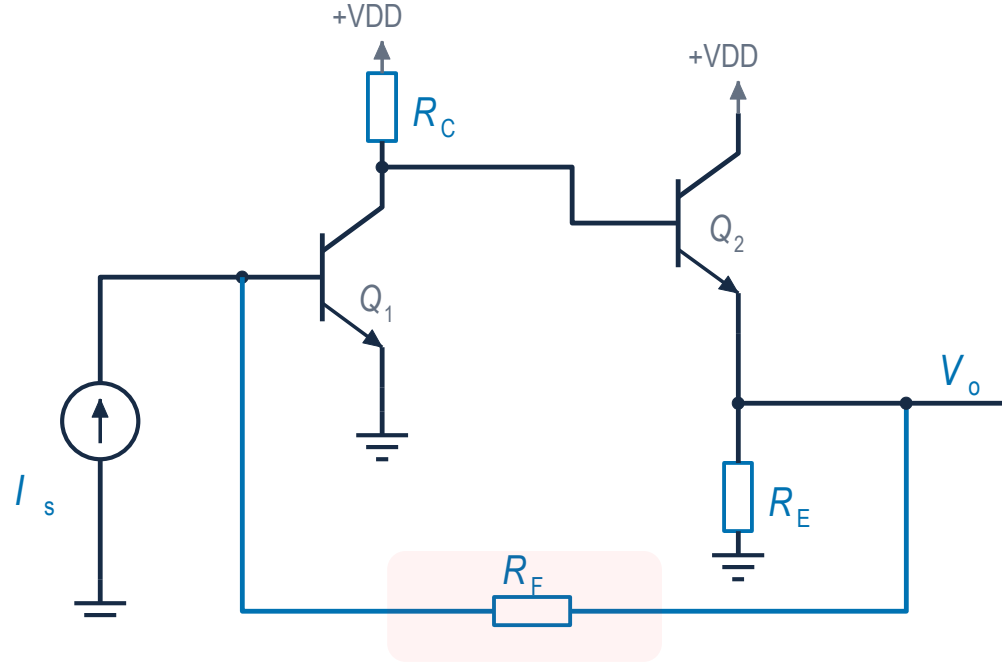
Transconductance amplifier circuit

Negative Feedback Topologies (8/8)



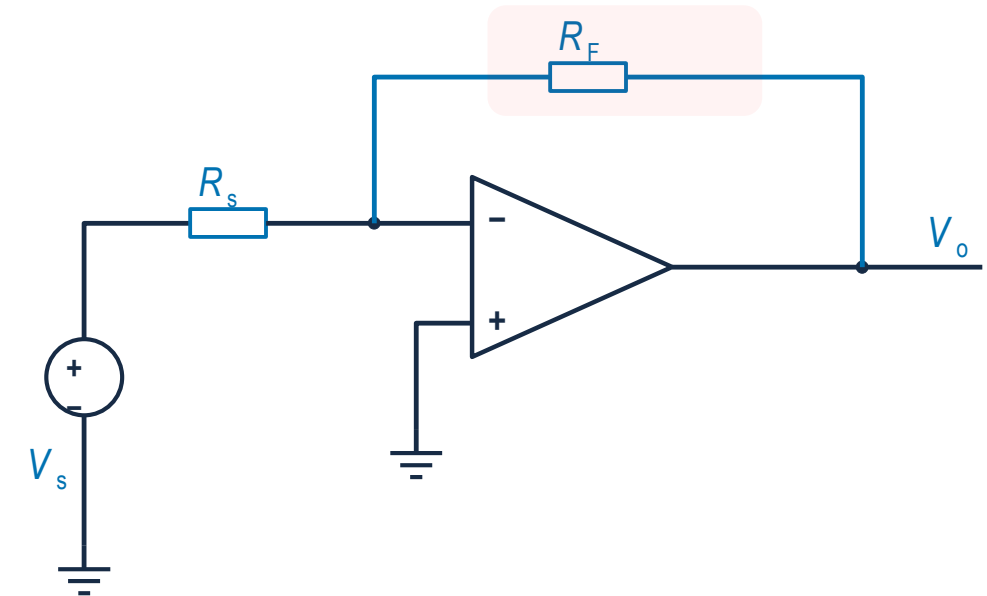
□ Circuit Examples

Shunt-Shunt



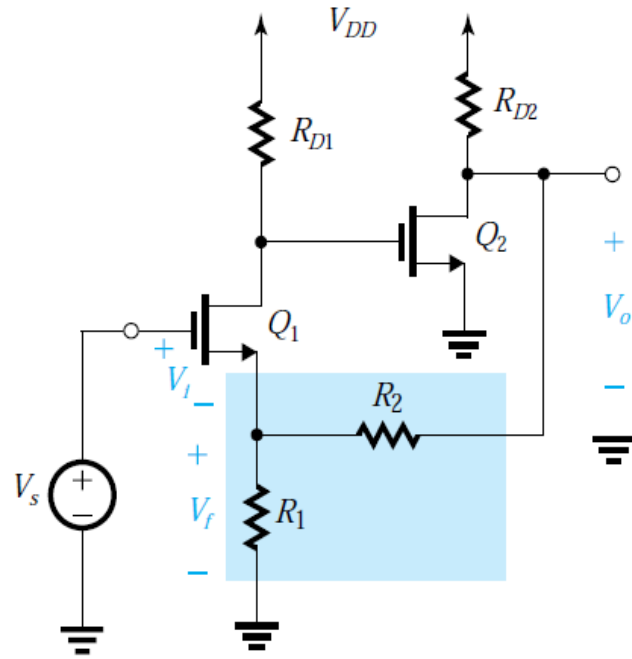
Two-stage BJT amplifier circuit

Shunt-Shunt

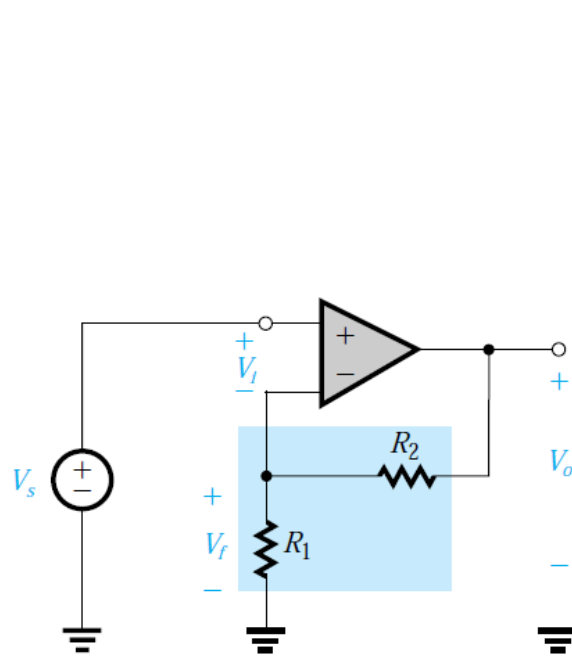


Inverting amplifier circuit

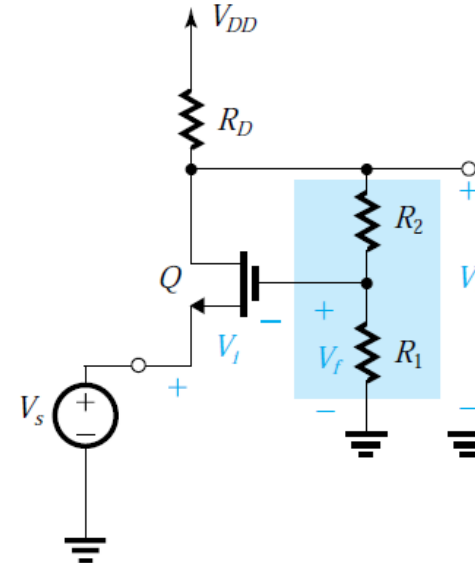
Concept check !



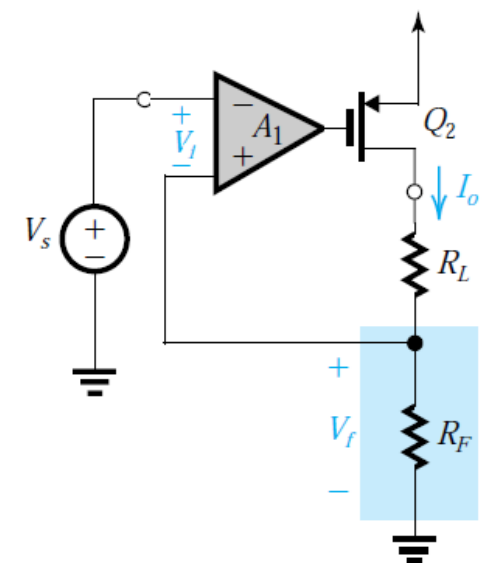
(a)



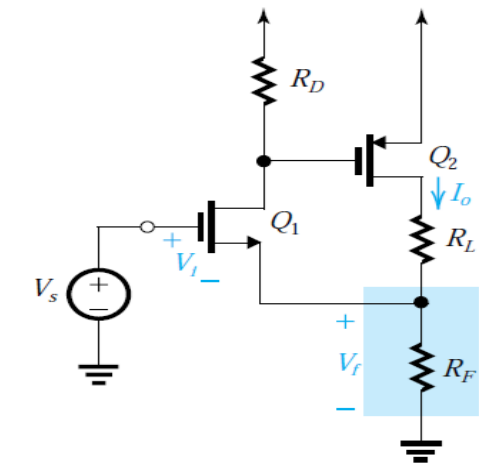
(b)



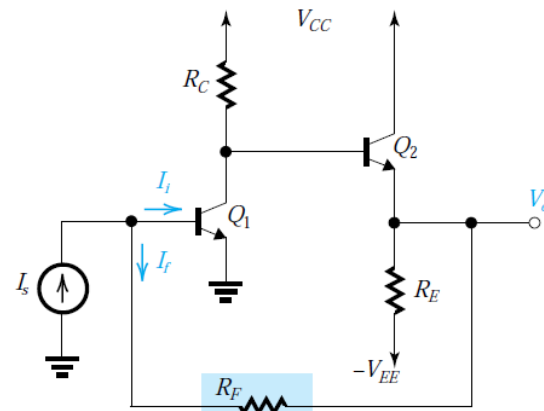
(c)



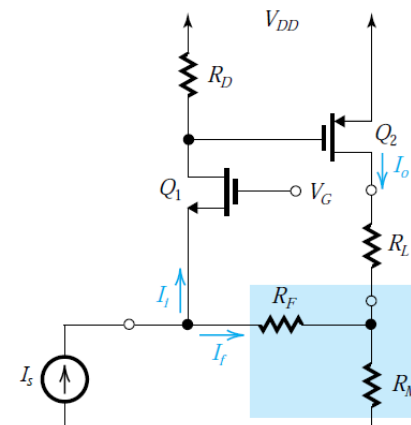
(d)



(e)



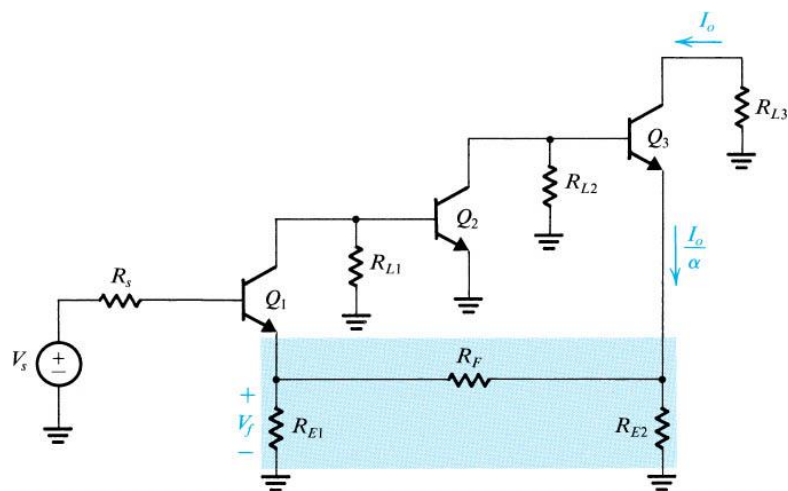
(f)



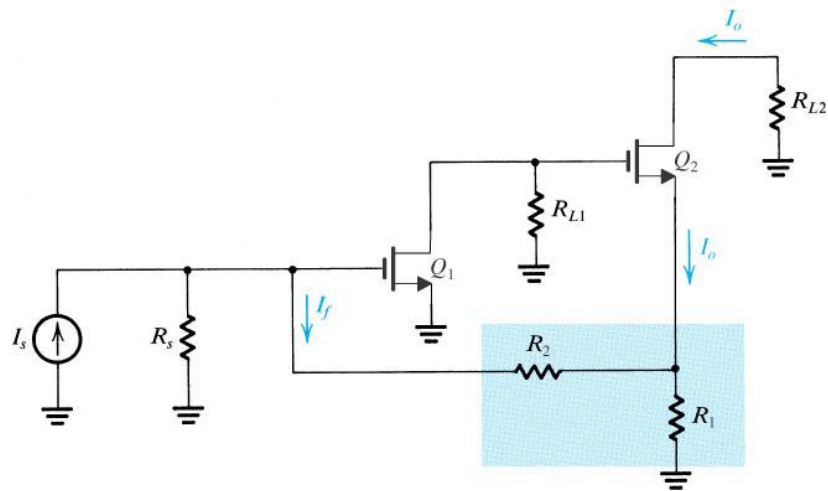
(g)



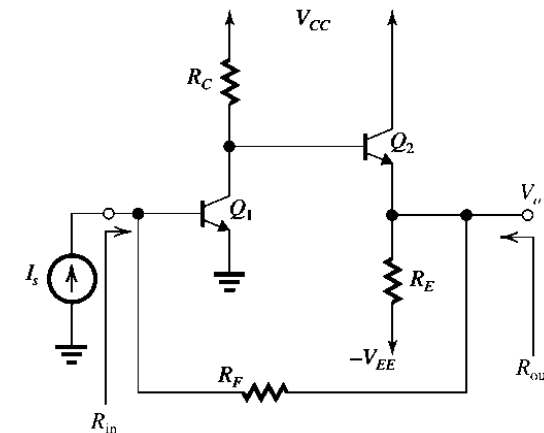
Concept check !



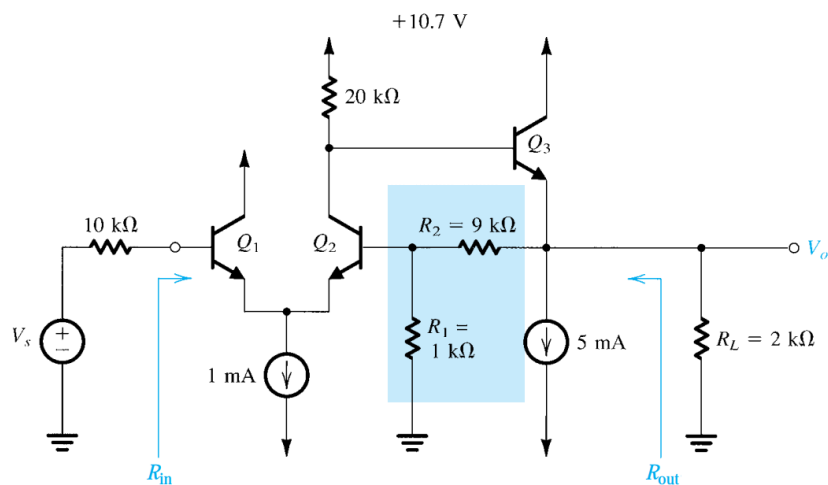
(a) Series-Series



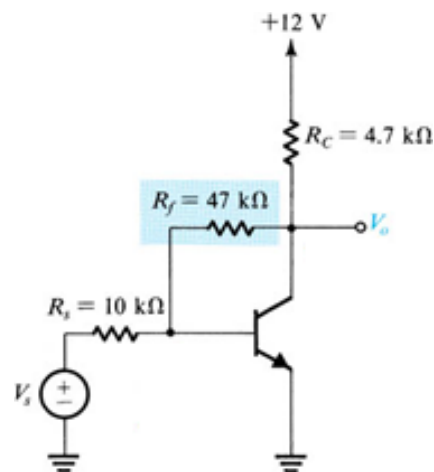
(b) Shunt-Series



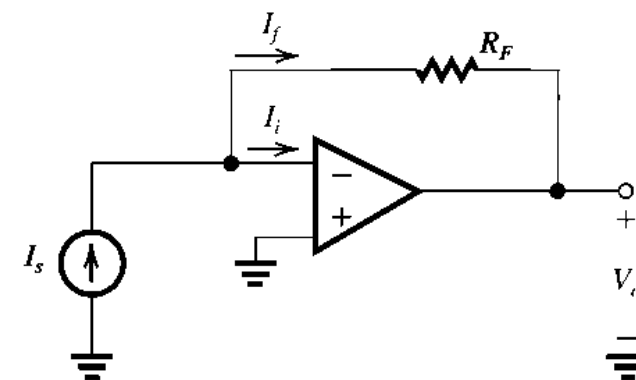
(c) Shunt-Shunt



(d) Series-Shunt



(e) Shunt-Shunt



(f) Shunt-Shunt



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Negative Feedback: Benefits and Tradeoffs



❑ Feedback exchanges closed-loop gain for more controlled amplifier behavior.

Benefits



1-Gain robustness

Less sensitivity to forward-gain variation.

2-Bandwidth extension

Extension within the assumed frequency model.

3-Internal disturbance/noise suppression

Effect depends on the noise injection point.

4-Lower in-loop distortion

Improved linearity within the output swing.

5-Impedance control

Resistance changes depend on topology.

Tradeoffs



1-Reduced gain

Lower gain from input to output.

2-Stability constraints

Loop dynamics require assessment.

3-Circuit complexity

Added networks, loading, and power.



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Gain Robustness (1/2)



□ How Negative Feedback Desensitizes Gain

1. Express the closed-loop gain

$$A_f = \frac{A}{1 + A\beta} = A(1 + A\beta)^{-1}$$

2. Differentiate with respect to A

$$\begin{aligned} \frac{dA_f}{dA} &= (1 + A\beta)^{-1} - A\beta(1 + A\beta)^{-2} \\ &= \frac{1 + A\beta - A\beta}{(1 + A\beta)^2} = \frac{1}{(1 + A\beta)^2} \end{aligned}$$

The feedback factor β is held constant.

3. Normalize the gain change

$$\frac{dA_f}{A_f} = \frac{1}{1 + A\beta} \cdot \frac{dA}{A}$$

The fractional gain variation is reduced by the factor

$$1 / (1 + A\beta).$$

For high loop gain

$$A\beta \gg 1$$

Changes in open-loop gain cause much smaller fractional changes in closed-loop gain.

Gain Robustness (2/2)



□ Gain Robustness: Numerical Comparison

Illustrative parameters: $\beta = 0.01$

Quantity	Initial value	Changed value
Forward gain A	1000	500
Loop gain $A\beta$	10	5
Closed-loop gain	90.91	83.33

A falls by 50%; the closed-loop gain falls by approximately 8.33%.





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Bandwidth Extension (1/3)



- ❑ For upper Cutoff, negative feedback reduces midband gain and raises the upper cutoff frequency

1. Open-loop response

$$A(s) = \frac{A_m}{1 + s/\omega_H}$$

2. Apply negative feedback

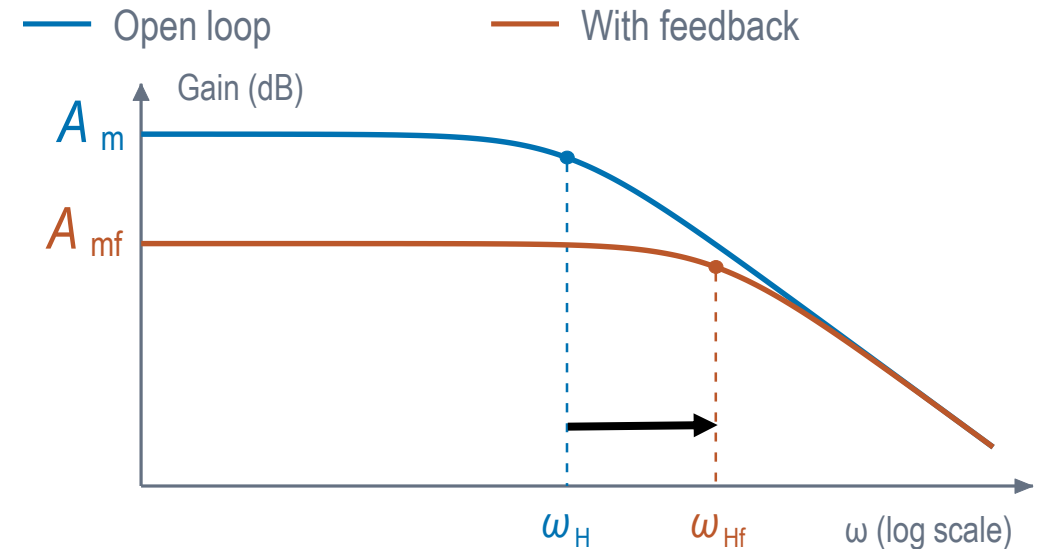
$$A_f(s) = \frac{A(s)}{1 + A(s)\beta} = \frac{\frac{A_m}{1 + s/\omega_H}}{1 + \frac{A_m}{1 + s/\omega_H} \beta}$$

3. Normalize the response

$$A_f(s) = \frac{A_m}{1 + A_m \beta + \frac{s}{\omega_H}} = \frac{A_m/(1 + A_m \beta)}{1 + \frac{s}{\omega_H(1 + A_m \beta)}}$$

$$A_f(s) = \frac{A_{mf}}{1 + s/\omega_{Hf}}$$

Magnitude response



$$A_{mf} = \frac{A_m}{1 + A_m \beta}$$

$$\omega_{Hf} = \omega_H(1 + A_m \beta)$$



Constant gain–bandwidth product

$$A_{mf} \omega_{Hf} = A_m \omega_H$$

Bandwidth Extension (2/3)



- ❑ For Lower Cutoff, negative feedback reduces gain and lowers the lower cutoff frequency.

1. Open-loop response

$$A(s) = \frac{A_m s}{s + \omega_L}$$

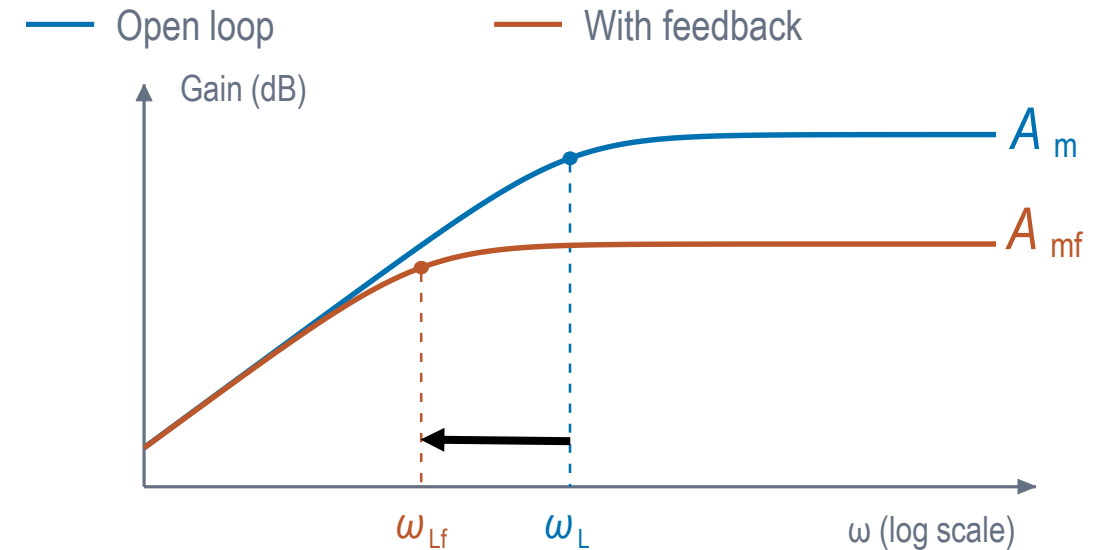
2. Apply negative feedback

$$A_f(s) = \frac{A(s)}{1 + A(s)\beta} = \frac{\frac{A_m s}{s + \omega_L}}{1 + \frac{A_m s}{s + \omega_L}\beta}$$

3. Normalize the response

$$\begin{aligned} A_f &= \frac{A_m s}{s + \omega_L + A_m s \beta} = \frac{A_m s}{\omega_L + (1 + A_m \beta)s} \\ &= \frac{\frac{A_m}{(1 + A_m \beta)} s}{s + \frac{\omega_L}{(1 + A_m \beta)}} = \frac{A_{mf} s}{s + \omega_{Lf}} \end{aligned}$$

Magnitude response



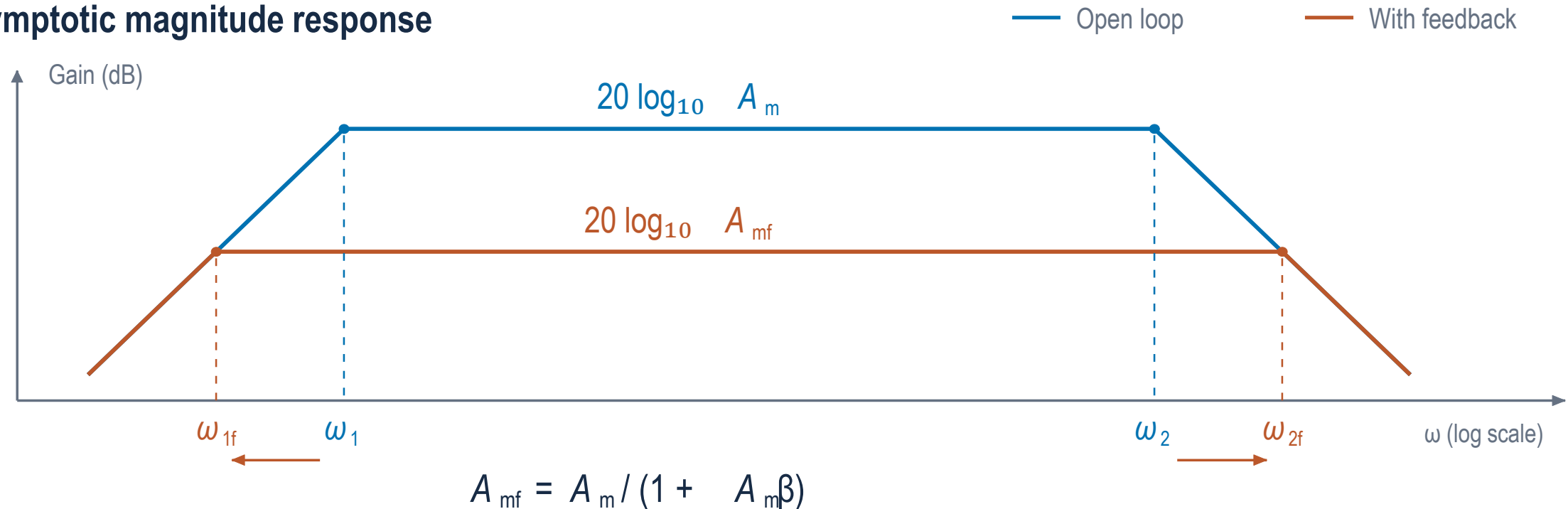
$$\begin{aligned} A_{mf} &= \frac{A_m}{1 + A_m \beta} \\ \omega_{Lf} &= \frac{\omega_L}{1 + A_m \beta} \end{aligned}$$

Bandwidth Extension (3/3)



- ❑ **Bandwidth Extension Across Both Cutoffs**
- ❑ Negative feedback lowers midband gain while widening the useful frequency range.

Asymptotic magnitude response



Lower cutoff decreases

$$\omega_{1f} = \frac{\omega_1}{1 + A_m \beta}$$

Upper cutoff increases

$$\omega_{2f} = \omega_2 (1 + A_m \beta)$$

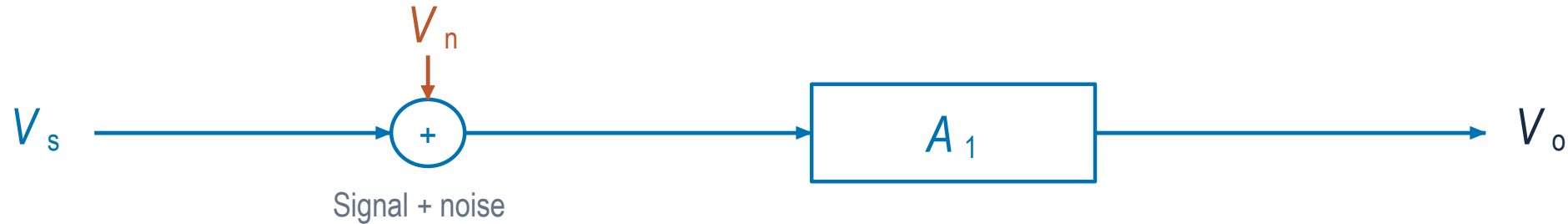


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Internal Disturbance/Noise Suppression



- ❑ Case 1: signal and noise enter the same amplifier input.



Output components

$$V_o = A_1 V_s + A_1 V_n$$

Signal contribution

Noise contribution

Signal-to-noise amplitude ratio

$$S/N = \frac{A_1 V_s}{A_1 V_n} = \frac{V_s}{V_n}$$

Interpretation

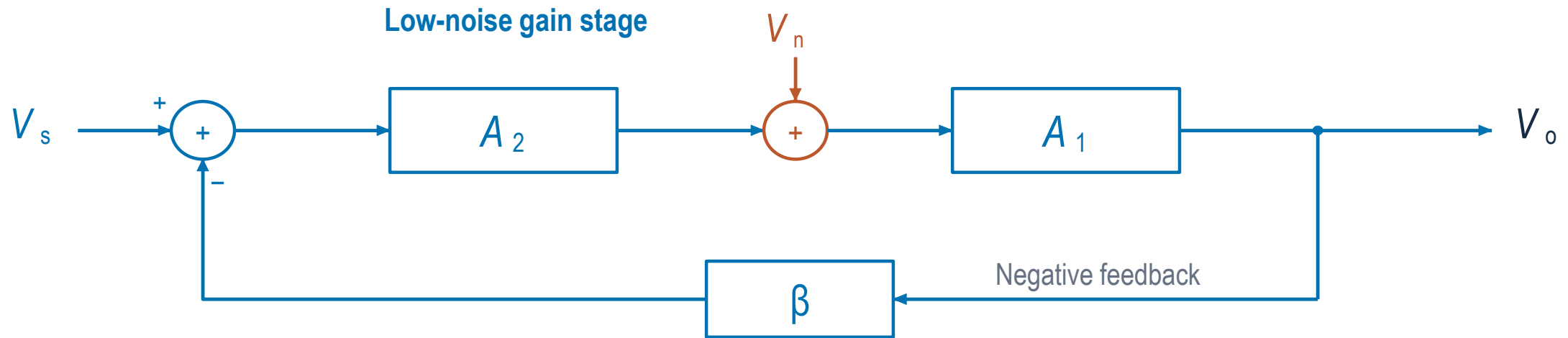
The amplifier applies the same gain to the input signal and input noise.

Increasing A_1 leaves their ratio unchanged.

Internal Disturbance/Noise Suppression



- ❑ Case 2: noise is introduced after a low-noise preamplifier inside the feedback loop.



Output signal and noise contributions

$$V_o = V_s \underbrace{\frac{A_1 A_2}{1 + A_1 A_2 \beta}}_{\text{Signal transfer}} + V_n \underbrace{\frac{A_1}{1 + A_1 A_2 \beta}}_{\text{Noise transfer}}$$

$$S/N = \frac{A_2 V_s}{V_n}$$



The S/N improvement comes from low-noise gain A_2 before noise injection.



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Lower in-loop distortion



- ❑ A lower closed-loop gain widens the input range before output saturation.

Without feedback

$$A = \frac{V_o}{V_i} = 1000$$

The open-loop gain is the transfer-curve slope.

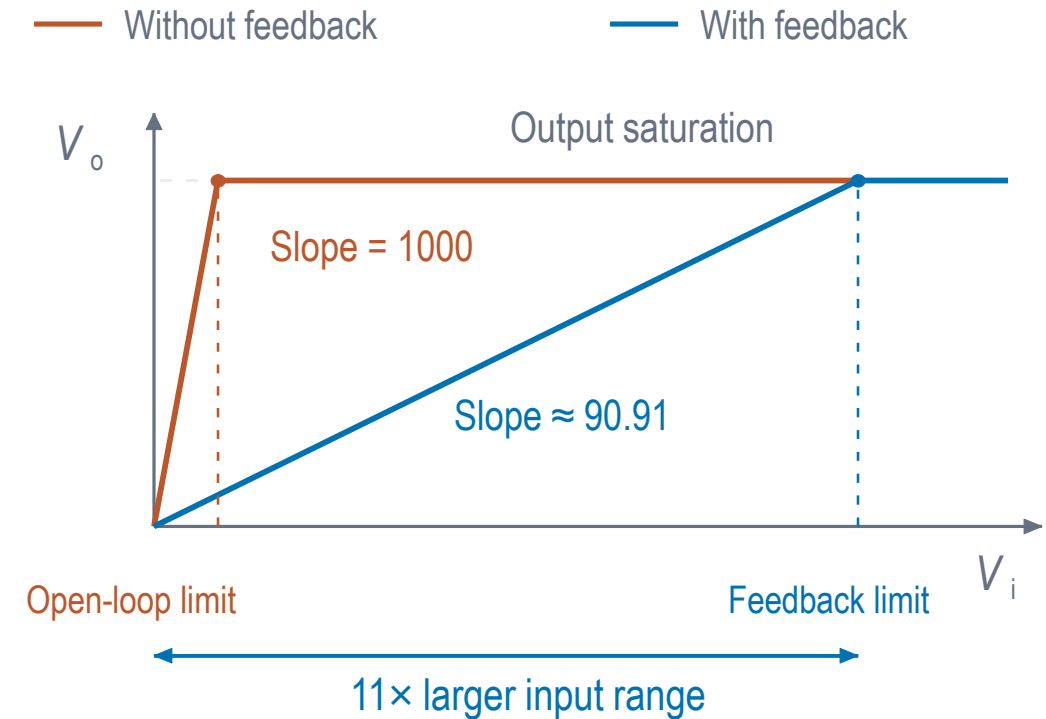
With negative feedback

$$\beta = 0.01 \quad A\beta = 10$$

$$A_f = \frac{A}{1 + A\beta}$$

$$A_f = \frac{1000}{11} \approx 90.91$$

Transfer characteristic



For the same output swing, negative feedback increases the input range by 11×.

Feedback suppresses in-loop distortion within the output swing; saturation still causes clipping.



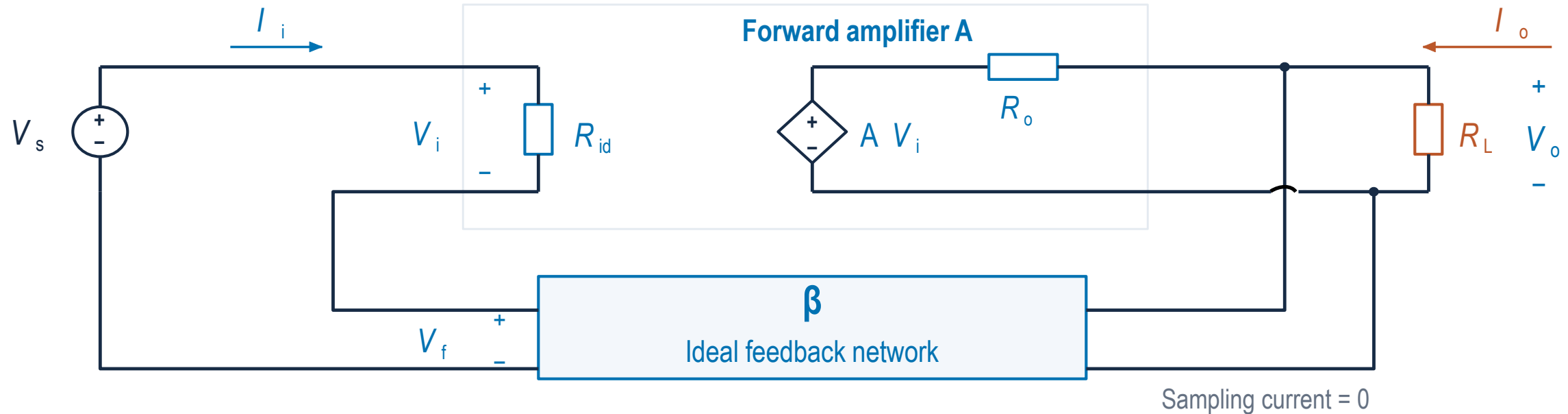
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Impedance control (1/2)



- ❑ **Series-Shunt Feedback** → Series voltage mixing raises input resistance; shunt voltage sampling lowers output resistance.

Ideal voltage-feedback model



Series mixing at the input

$$R_{if} = R_{id} (1 + A\beta)$$

Shunt sampling at the output

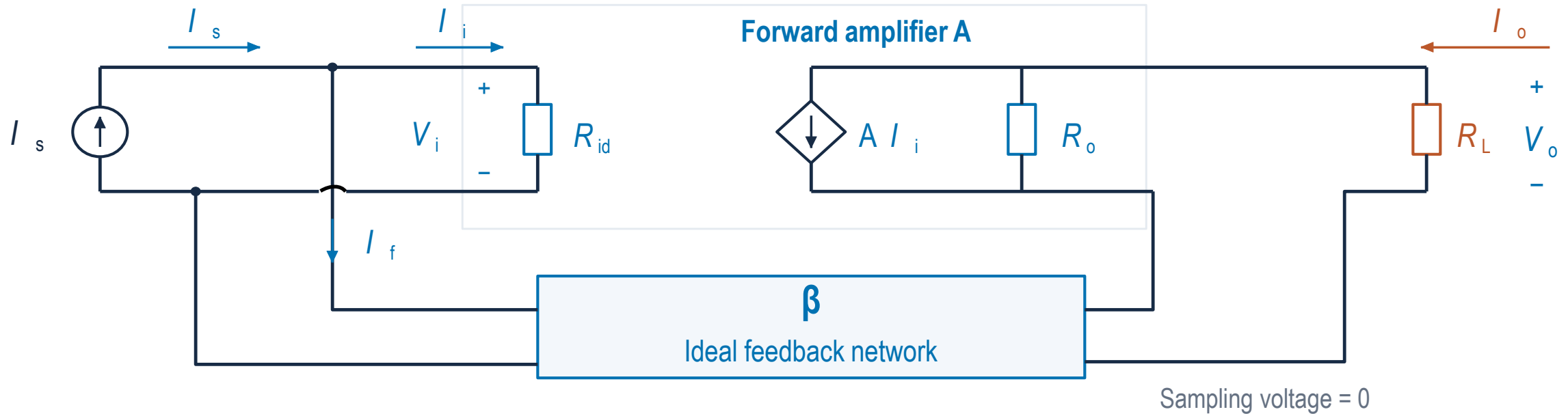
$$R_{of} = \frac{R_o}{1 + A\beta}$$

Impedance control (2/2)



- ❑ **Shunt-Series Feedback** → Shunt current mixing lowers input resistance; series current sampling raises output resistance.

Ideal current-feedback model



Shunt mixing at the input

$$R_{if} = \frac{R_{id}}{1 + A\beta}$$

Series sampling at the output

$$R_{of} = R_o(1 + A\beta)$$



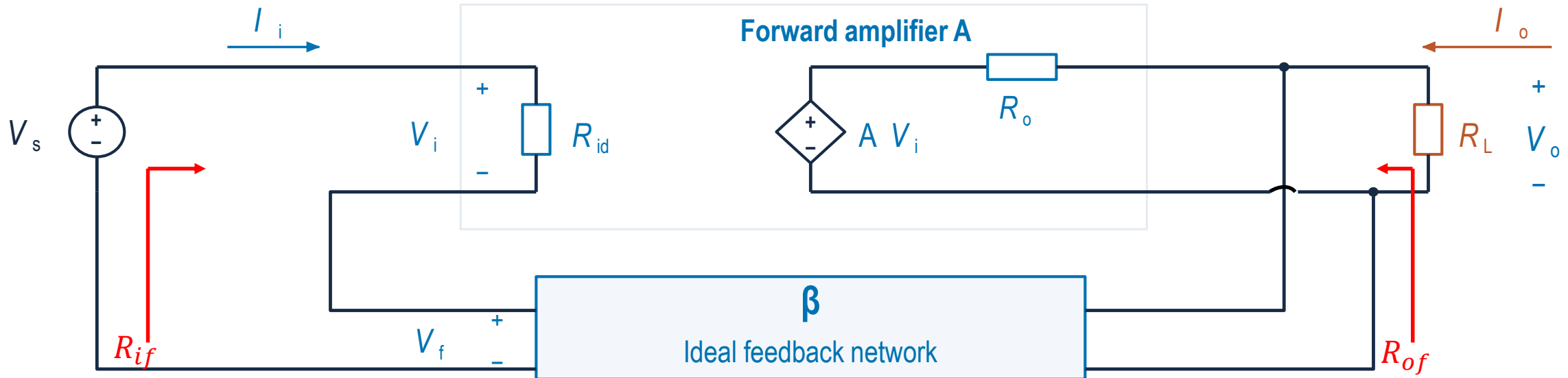
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Resistance Derivations (1/4)



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Ideal voltage-feedback model



Series mixing at the input

$$R_{if} = R_{id} (1 + A\beta)$$

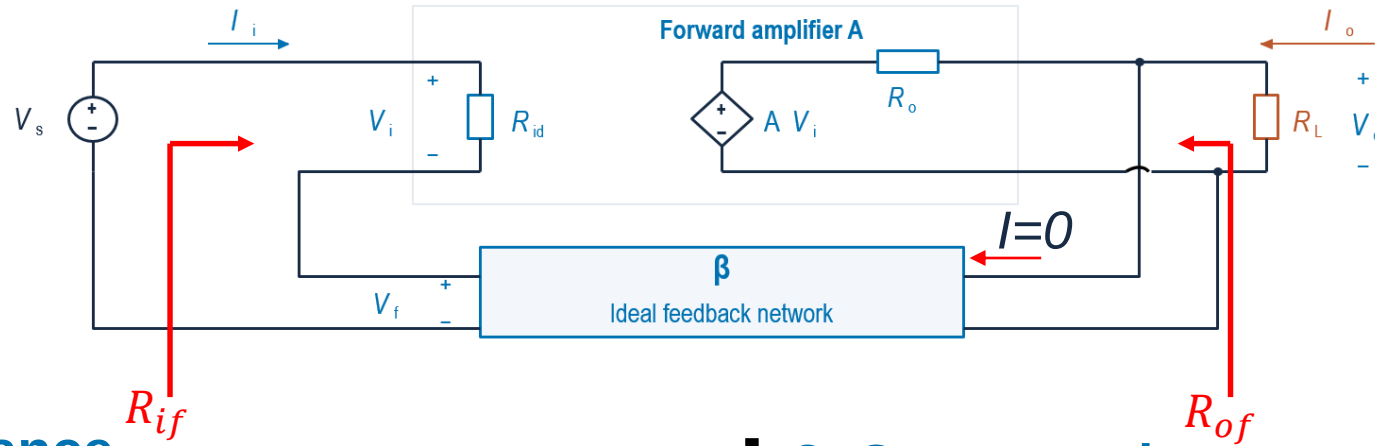
Shunt sampling at the output

$$R_{of} = \frac{R_o}{1 + A\beta}$$

Resistance Derivations (2/4)



Topology: Series-Shunt



1. Input resistance

Open-circuit output: $R_L \rightarrow \infty$

$$R_{if} = \frac{V_s}{I_i} \quad \& \quad I_i = \frac{V_i}{R_{id}} \quad \gggg \quad R_{if} = \frac{V_s}{V_i} R_{id}$$

$$V_s = V_i + V_f = V_i + \beta V_o$$

$$V_o = A V_i$$

$$V_s = V_i (1 + A\beta)$$

$$R_{if} = R_{id} (1 + A\beta)$$



$$\gggg \quad \frac{V_s}{V_i} = (1 + A\beta)$$

2. Output resistance

Set $V_s = 0$ and apply an output test current.

$$R_{of} = \frac{V_o}{I_o} \quad \text{with } I=0$$

$$I_o = \frac{V_o - A V_i}{R_o}$$

$$V_s = V_i + V_f = 0 \quad \gggg \quad V_i = -V_f$$

$$I_o = \frac{V_o + A V_f}{R_o} \quad \& \quad V_f = \beta V_o$$

$$I_o = \frac{V_o (1 + A\beta)}{R_o} \quad \gggg \quad R_{of} = \frac{R_o}{1 + A\beta}$$

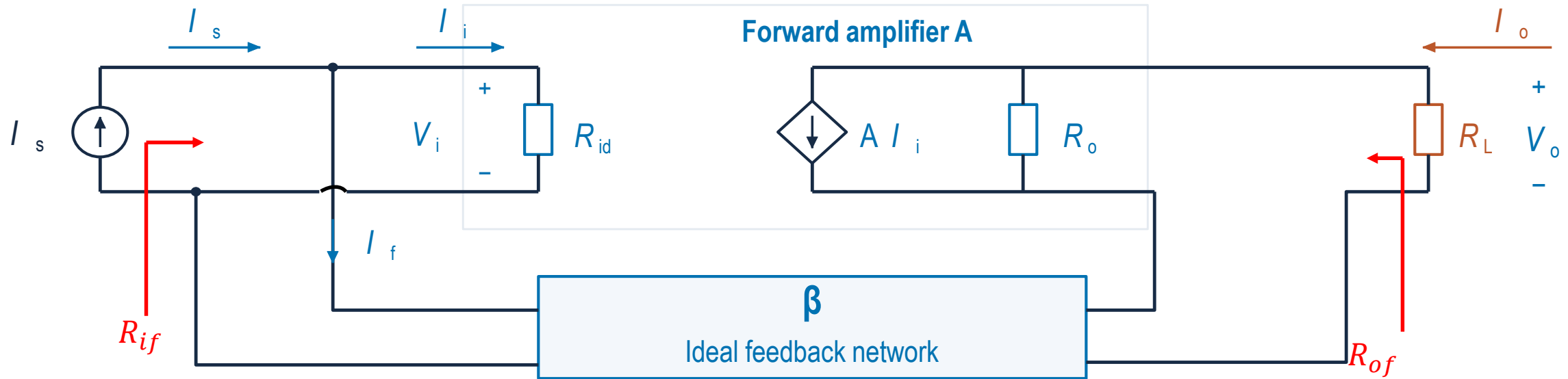


Resistance Derivations (3/4)



- ❑ **Shunt-Series Feedback** → Shunt current mixing lowers input resistance; series current sampling raises output resistance.

Ideal current-feedback model



Shunt mixing at the input

$$R_{if} = \frac{R_{id}}{1 + A\beta}$$

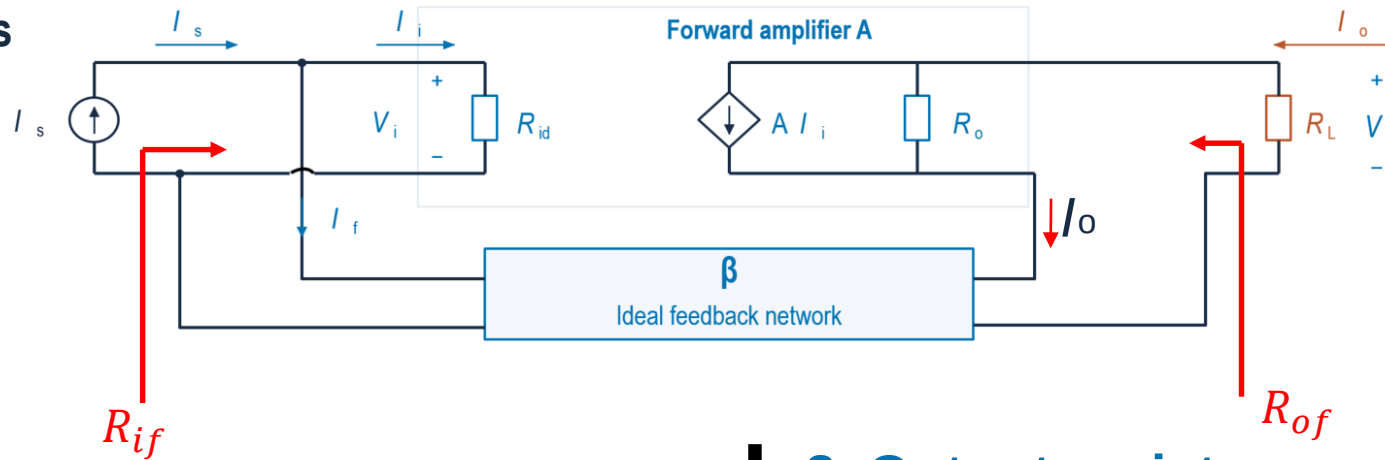
Series sampling at the output

$$R_{of} = R_o(1 + A\beta)$$

Resistance Derivations (4/4)



Topology: Shunt – Series



1. Input resistance

Short-circuit output: $R_L \rightarrow 0$

$$R_{if} = \frac{V_i}{I_s}$$

$$I_s = I_i + I_f = \frac{V_i}{R_{id}} + \beta I_o \quad \& \quad I_o = A I_i$$

$$I_i = \frac{V_i}{R_{id}}$$

$$I_s = \frac{V_i}{R_{id}} + \beta A I_i$$

$$I_s = \frac{V_i}{R_{id}} + \beta A \frac{V_i}{R_{id}} \gg R = \frac{R_{id}}{1 + A\beta}$$



2. Output resistance

Set $I_s = 0$ and apply an output test current.

$$R_{of} = \frac{V_o}{I_o}$$

$$I_o = \frac{V_o}{R_o} + A I_i \quad \& \quad I_s = I_i + I_f = 0$$

$$I_i = -I_f = -\beta I_o$$

$$I_o = \frac{V_o}{R_o} - A\beta I_o$$

$$I_o(1 + A\beta) = \frac{V_o}{R_o} \gg R_{of} = R_o(1 + A\beta)$$



Thanks!

